
EE 606: Introduction to High Voltage Engineering and Dielectric Breakdown

Course Code	EE 606
Course Name	Introduction to High Voltage Engineering and Dielectric Breakdown
L-T-P-C	3-0-0-3
Prerequisites	EE 303 Power Systems or teachers consent
Intended for	B.Tech./M.S./Ph.D.
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- Electric Field Strength. [4 Lectures]
- Gaseous Dielectrics. [9 Lectures]
- Properties of Liquid and Solid Dielectrics. [6 Lectures]
- Breakdown in Liquid and Solid Dielectrics. [4 Lectures]
- Generation of High Test Voltages. [6 Lectures]
- Measurement of High Test Voltages. [4 Lectures]
- Non-destructive High Voltage Testing and Quality Control. [4 Lectures]
- Insulation Coordination and Over Voltages in Power Systems. [2 Lectures]
- Introduction National and International standards such as IEC-60060-1, 60-2, etc. [1 Lecture]
- Recent trends and developments. [2 Lectures]

Text Books

1. R. Arora, W. Mosch, **High Voltage and Electrical Insulation Engineering**, IEEE Press, 2011.
2. M. S. Naidu, Kamaraju, **High Voltage Engineering**, TMH, 2009.

References

1. Kuffel, E., **High voltage engineering**, Newnes, 2009.
2. Alston L. L., **High Voltage Technology**, Oxford University Press, 2011.

Laboratory Modules

1. To Study the corona phenomena alongside thin copper wire by use of power frequency HV test source.
2. To study electrical breakdown of a sample of in-service HV insulating oil for electrical breakdown against contamination & moisture.
3. To carry out measurements of earthling resistance by three probe method.
4. To carry out measurement of earth resistance by four probe method.
5. To carry out air insulation breakdown studies by using uniform sphere gap spacing.
6. To carry out air insulation breakdown studies by using non-uniform sphere gap spacing.
7. To carry out electrical breakdown studies on solid insulations using test setup.